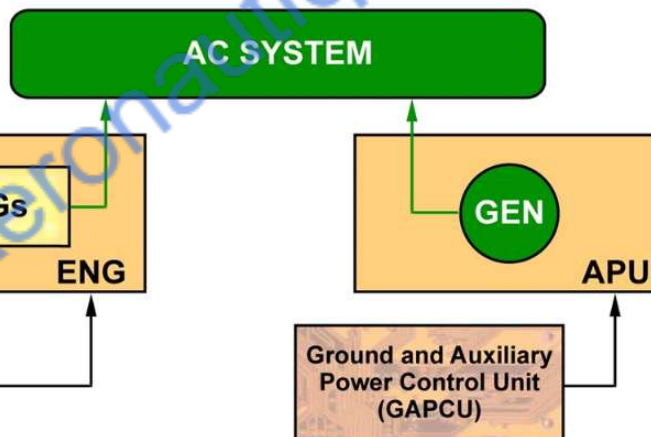


ETUDES SYSTEMES

ATA 24

PART 1



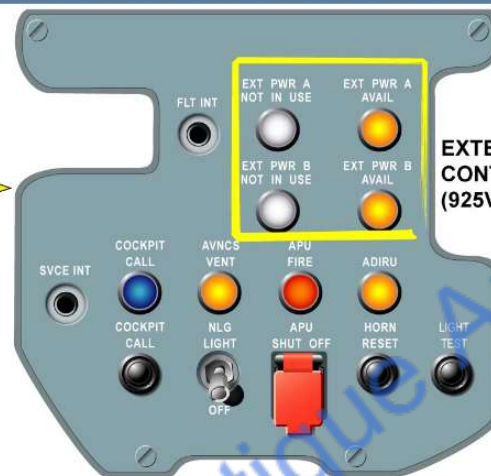
AC GENERATION - IDG (b)

In flight the AC electrical system is supplied by engine driven Integrated Drive Generators (IDGs). There is one IDG per engine, directly connected on the related engine gearbox pad. Each IDG is controlled and monitored by its related Generator Control Unit (GCU).

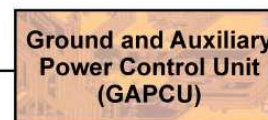
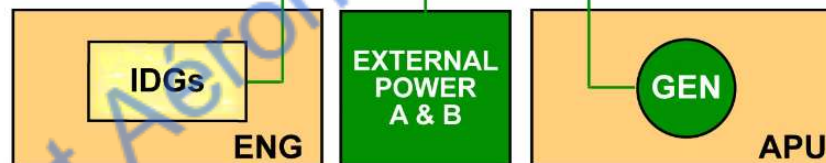
IDG: Integrated Drive Generator
GEN: Generator



NOSE LANDING GEAR (NLG)



**EXTERNAL POWER
CONTROL PANEL
(925VU)**



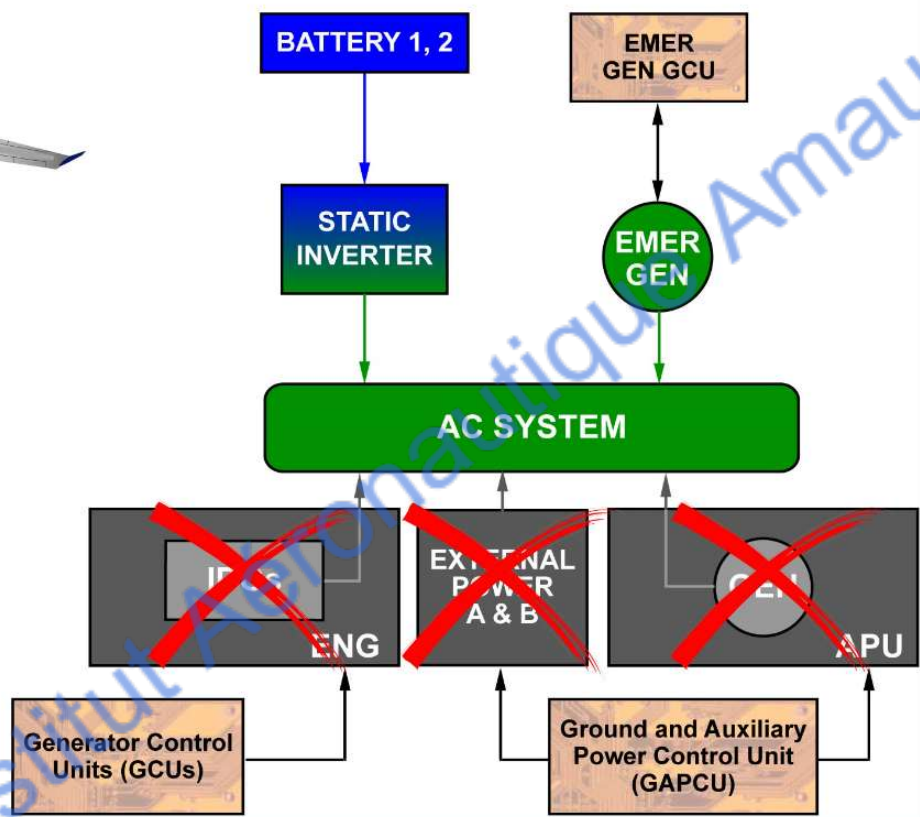
AC GENERATION - EXTERNAL POWER

On ground, the entire AC electrical system can also be supplied by two external power sources called:

- EXTERNAL PoWeR A,
- and/or EXT PWR B.

The external power control panel gives access to two EXT PWR receptacles for connection of the ground power

IDG: Integrated Drive Generator
GEN: Generator



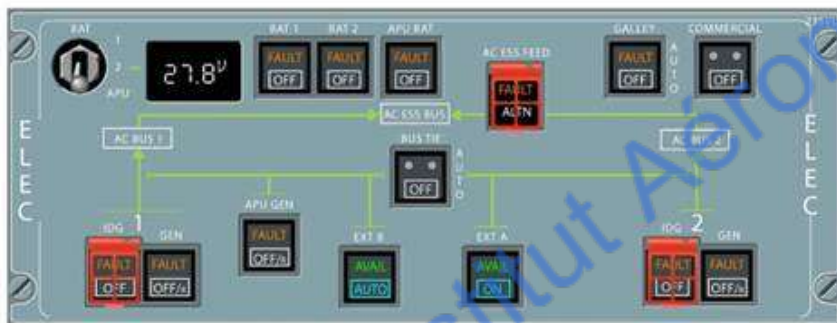
AC GENERATION - EMERGENCY
controlled and monitored by the CSM/G Generator Control Unit (also called EMER GCU). If no other AC power source is available (IDGs, APU GEN, EXT PWR and EMER GEN), a static inverter, supplied by the batteries, can supply a part of AC system called AC Essential electrical system network.

IDG: Integrated Drive Generator
GEN: Generator

IAAG - Institut Aeronautique Amaury de la Grange



ECAM PAGE

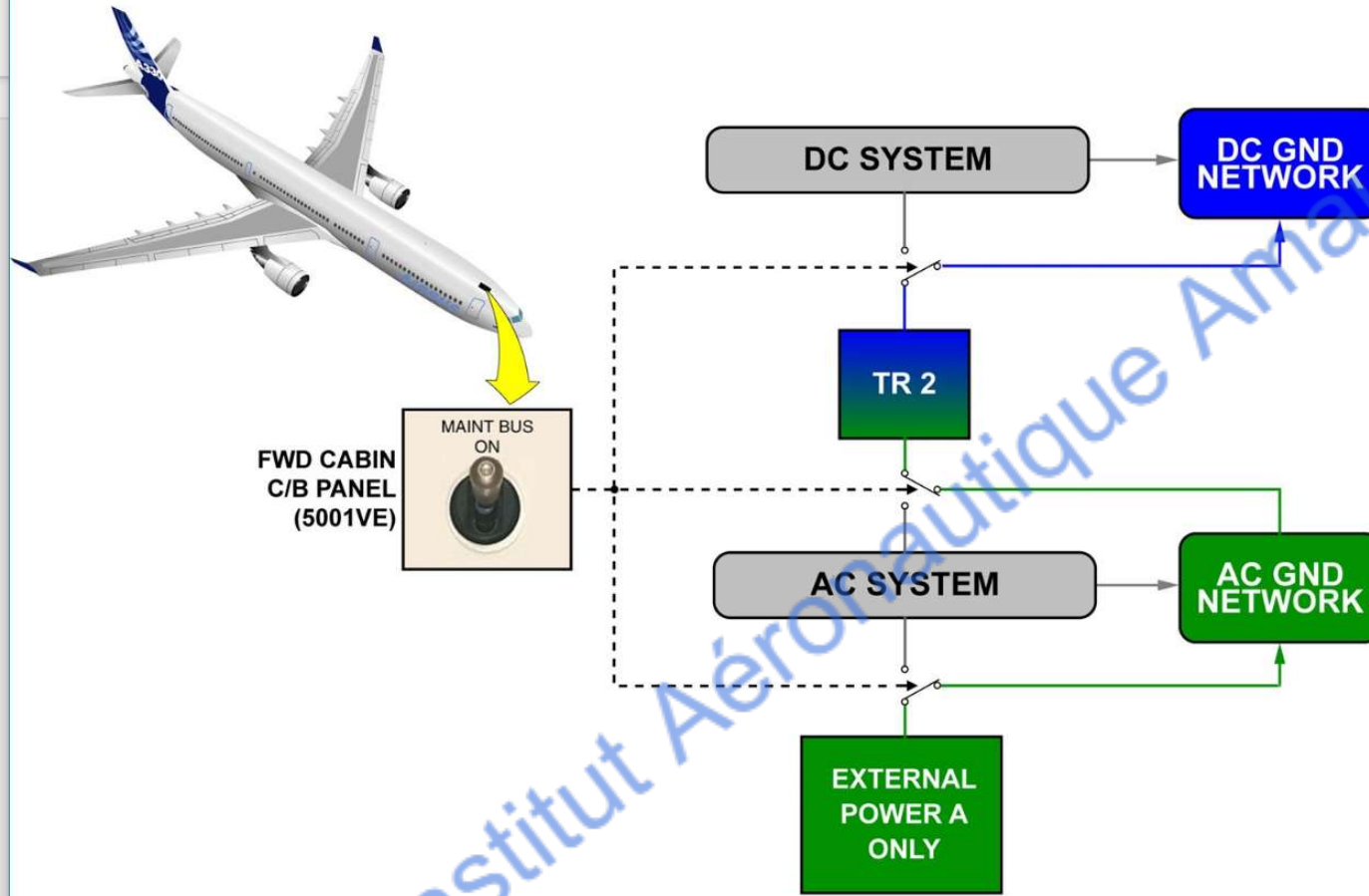


(235VU)



(211VU)

CONTROL & INDICATING - AC GENERATION panel contains the P/BSWs and indicators to control and monitor the emergency power source, which is the emergency generator (CSM/G). The AC electrical page of the ECAM shows the AC sources and how they are connected to the network. The most important parameters are also monitored.



GROUND SERVICE NETWORK

A MAINTenance BUS switch is used to supply the ground service electrical network without energizing the whole electrical network. When the switch is in the ON position, the EXT PWR A only, supplies the ground service network.

**Circuit
Breaker
Monitoring
Unit
(CBMU)**



**CIRCUIT BREAKERS (C/Bs)
(AVIONIC COMPARTMENT)**

**AC & DC
SYSTEM**

**AC & DC
CONTACTORS**

**Electrical
Contactor
Management
Units
(ECMU)**

**DC LOAD
DISTRIBUTION**

**AC LOAD
DISTRIBUTION**

CONTACTOR MANAGEMENT AND C/Bs MOI

Contactor Management Units (ECMUs).
A specific computer named Circuit Breaker Monitoring Unit (CBMU) monitors the C/Bs and the RCCBs status. It indicates on the ECAM any tripped or manually pulled monitored C/Bs or RCCBs. The monitored C/Bs are usually green color.

C/B 1/1

EXTC FAN BULK CRG	A37	5HN
ECAM SWTG DMC 1	B75	11WT1
303 PP	E24	9PB
PACK 2 L2 115VAC	E44	3H42
ESS BUS NORM SPLY	A37	1XC
PCK 2 FLOW CTL	F80	1HB2
PACK 2 L1 115VAC	H48	2WC
EXT HORN	H78	11WT1
AVNCS EXTRACT FAN	J2	9EP
ADIRU 1 & 3 SWTG	K2	11WT1
AEVC	L5	1CW
SFCC 1 SLAT	L62	4HG
ISOL VLV LDWCR	L70	1HR
CARGO VENT CTL L2	L72	11WT1
501 PP ESS/GND SWTG	N42	11WT1
CAB PRESS MAN	P11	8PR
PHC 1	P12	11WT1

TAT	+29 °C	GW	240000 KG
SAT	+29 °C	17 H 05	GWCG 28.2 %
ISA	+5 °C		

C/B ECAM PAGE